Chapter 5:

Divisibility: If $a, b \in \mathbb{Z}$, then we say a divides b and we write $\boxed{a|b}$ if there exists $c \in \mathbb{Z}$ such that $a \cdot c = b$

$\boxed{\text{This is NOT } a/b}$

Ex. we see $5|10$ "5 divides 10" since $5 \cdot 2 = 10$

we say 10 is divisible by 5

the building block of + in $\mathbb{N}$ is successor (σ) and (1)

the building block of $\cdot$ is prime ~~#s~~.

1, 2, 3, 4 (2·2), 5, (6) (2·3), 7 .....

Numbers break into primes

The Quotient - Remainder Theorem

If $a, b \in \mathbb{Z}$ with $a \neq 0$, then we can write $b = a \cdot q + r$ for a unique $q, r \in \mathbb{Z}$ with $0 \leq r \leq a$

quotient

Remainder

Ex: $a = 5$, $b = 10$ then $10 = 5 \cdot 2 + 0$

$a = 13$ $b = 141$ $141 = 13 \cdot 10 + 11$ ✓ $0 \leq 11 \leq 13$

Proof: If $b < a$, then $b = a \cdot 0 + b$.
Since $0 \le b < a$ this works great.
If $b = a$, $b = a \cdot 1 + 0$.

Suppose now that $b > a$. Consider the set

$S = \{b - xa : x \in \mathbb{N} \text{ and } b - x \cdot a \ge 0\}$

Note $S \subset \mathbb{N}$ and S is non-empty ($S \neq \emptyset$) since $b - a \in S$.
By the well-ordering principle S has a least element.
Call it r_0 Note $r_0 = b - x_0 a$ for some $x_0 \in \mathbb{N}$
We claim $0 \le r_0 < a$. For a contradiction suppose $r_0 \ge a$.
Thus $r_0 - a \ge 0$. We compute $r_0 - a = b - x_0 a - a = b - (x_0 + 1)a$
But $b - (x_0 + 1)a \in S$, and contradicts r_0 being the least element of S. Therefore $0 \le r_0 < a$ as desired. We conclude since $r_0 = b - x_0 a$, that

$b = x_0 \cdot a + r_0$ with $0 \le r_0 < a$.

Sketch of uniqueness: $b = aq_1 + r_1 = aq_2 + r_2$

$0 \le r_1, r_2 < a$. Then $aq_1 - aq_2 = r_2 - r_1$

So $a(q_1 - q_2) = r_2 - r_1 \longrightarrow$ Thus, a divides $r_2 - r_1$

Since r_2 and r_1 are both $< a$ and $\ge a$ we see

$r_2 - r_1 = 0$ so $r_2 = r_1$

Euclid's Idea: Do quotient remainder Thm over and over. Then, we can find the greatest common divisor of a and b.

Euclidean Algorithm: Given $a, b \in \mathbb{Z}$ we will find the gcd of a and b. How?

Q.R. Thm:

$b = a \cdot q + r \quad 0 \le r < a$

$\gcd(788, 134)$

$788 = 134 \cdot 5 + 118.$

Now $a = rq_1 + r_1 \quad 0 \le r_1 < r \qquad 134 = 118 \cdot 1 + 16$

$r = r_1 q_2 + r_2 \quad 0 \le r_2 < r_1 \qquad 118 = 16 \cdot 7 + 6$

$r_1 = r_2 q_3 + r_3 \quad 0 \le r_3 < r_2 \qquad 16 = 6 \cdot 2 + 4$

$r_2 = r_3 q_4 + r_4 \quad 0 \le r_4 < r_3 \qquad 6 = 4 \cdot 1 + 2$

$r_3 = r_4 \cdot q_5 + r_5 \quad 0 \le r_5 < r_4 \qquad 4 = 2 \cdot 2 + 0$

The last non-zero remainder is the gcd. So 2

Bezout's Theory says you can work backwards:

So...

$118 = 788 - 134 \cdot 5$

$16 = 134 - 118 \cdot 1$

$6 = 118 - 16 \cdot 7$

$4 = 16 - 6 \cdot 2$

$2 = 6 - 4 \cdot 1$

$2 = 6 - (16 - 6 \cdot 2) \cdot 1$

$2 = 6 - 16 + 2 \cdot 6 = 3 \cdot 6 - 16$

$2 = 3(118 - 16 \cdot 7) - 16$

$2 = 3 \cdot 118 - 21 \cdot 16 - 16 = 3 \cdot 118 + 22 \cdot 16$

$2 = 3 \cdot 118 - 22(134 - 118)$

$2 = 25 \cdot 118 - 22 \cdot 134$

$2 = 25(788 - 134 \cdot 5) - 22 \cdot 13$

$2 = 25 \cdot 788 - 147 \cdot 134$

Do quotient-remainder common

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Q.R

Euclid's Algorithm for GCDs and Bézout's Identity

Number Systems

Theorem 28. (The Quotient-Remainder Theorem) *Let $a, b \in \mathbb{Z}$ with $a \neq 0$. There are unique $q, r \in \mathbb{Z}$ such that*

$$b = qa + r \quad \textit{with} \quad 0 \leq r < |a|.$$

If $r = 0$, then we say that a *divides* b and we write $a \mid b$. **This is not the same as a/b!!!** We define the *greatest common divisor* of two integers m and n, abbreviated $\gcd(m, n)$, to be the largest (in terms of absolute value) common divisor of m and n. For example, 12 and 18 have common divisors 1, 2, 3, and 6. Thus $\gcd(18, 12) = 6$.

By repeating the quotient-remainder theorem with a taking the place of b and r taking the place of a, we can compute the gcd of a and b. The algorithm of repeating the quotient remainder theorem is called the *Euclidean algorithm*.

If we work back through a completed Euclidean algorithm computation, we can find integers x and y such that $ax + by = \gcd(a, b)$. This result is often called Bézout's identity:

Bézout's Identity. *If $a, b \in \mathbb{Z}$, then there exist $x, y \in \mathbb{Z}$ such that $ax + by = \gcd(a, b)$.*

(1) Compute $\gcd(267, 112)$ using the Euclidean algorithm.

$267 = 112 \cdot 2 + 43.$

$112 = 43 \cdot 2 + 26$

$43 = 26 \cdot 1 + 17$

$26 = 17 \cdot 1 + 9$

$17 = 9 \cdot 1 + 8$

$9 = 8 \cdot 1 + 1$

$8 = 1 \cdot 8 + 0$

$GCD = 1$

(2) Working backwards from the last non-zero remainder, write $\gcd(267, 112)$ as a linear combination of 267 and 112. That is, find x and y so that $267x + 112y = \gcd(267, 112)$.